

ORIGINAL RESEARCH

Existing research guidelines for inclusive trials methodology. Working toward the integration of qualitative research, equity, diversity, and inclusion and trials methodology: a focused mapping review

Teena Thomas^a, Laura Jones^b, Shoba Dawson^{c,d}, Catherine Houghton^{a,e}, Andrew Hunter^{a,e}, Niamh O'Connell^a, Linda Biesty^{a,e,*}

^a*School of Nursing and Midwifery, University of Galway, Galway, Ireland*

^b*School of Health Sciences, University of Birmingham, Birmingham, UK*

^c*School of Medicine and Population Health, University of Sheffield, Sheffield, UK*

^d*Trial Forge Inclusive Research Centre (Sheffield), Sheffield, UK*

^e*Qualitative Research in Trials Centre (QUESTS), Heath Research Board Trials Methodology Research Network (HRB-TMRN), School of Nursing and Midwifery, University of Galway, Galway, Ireland*

Accepted 18 February 2026; Published online 21 February 2026

Abstract

Background: Equity, diversity, and inclusion (EDI) are increasingly emphasized in clinical research, yet practical guidance for applying inclusion principles across all clinical trial stages remain limited. Qualitative research can help address this gap by exploring participant experiences and identifying barriers to inclusive trial conduct. The aim of this review is to map guidelines and recommendations in relation to inclusive trials methodology and to identify areas across these resources that can be potentially informed by qualitative research. This review presents Phases 1 and 2 of the QuAlitative reSearch Supporting IncluSive Trials (Q-ASSIST) study.

Methods: In Phase 1, we conducted a focused mapping review and synthesis (FMRS) of publicly available guidelines related to Phase 3 and later clinical trials involving human participants. Eligible guidelines included those offering recommendations to enhance inclusive trial design and conduct. Each guideline was mapped to a priori data extraction framework informed by the Standard Protocol Items: Recommendations for Interventional Trials and Consolidated Standards of Reporting Trials 2025 guidelines. We then performed a narrative synthesis to examine the EDI focus of the selected guidelines and structured the findings according to the PRO-EDI Framework. In Phase 2, we juxtaposed the trial stages identified through our FMRS with O'Cathain et al's framework of qualitative research in trials.

Results: We identified 15 guidelines through FMRS. Through conceptual mapping to Standard Protocol Items: Recommendations for Interventional Trials and Consolidated Standards of Reporting Trials headings, we developed a 12-stage trial lifecycle framework (the Q-ASSIST trial stages model) to organize inclusion guidance. Most guidelines emphasized early stages of trials, with later stages less frequently addressed. Mapping to the PRO-EDI framework showed strong attention to race/ethnicity, gender, socioeconomic status, and disability, but limited focus on sexual identity, education, or intersectionality. Juxtaposition with O'Cathain et al's framework highlighted similar gaps in how qualitative research has been used, especially in later trial stages.

Conclusion: Current inclusion guidance is concentrated in the early stages of trial design, with limited attention to later trial stages. Qualitative research offers a valuable way to address these gaps by capturing participant perspectives and supporting inclusive practices

Funding: This study is funded by the Health Research Board— Trials Methodology Research Network in Ireland (grant ref: HRB-TMRN-2021-001). The funder did not participate in the design, analysis, or decision to publish this study.

Reflexivity statement: The authorship team brings diverse disciplinary backgrounds, methodological expertise, and lived experiences that shaped the focus, interpretation, and synthesis of this work. Team members include researchers with expertise in qualitative and mixed-methods research, trials methodology research, evidence synthesis around recruitment to trials, women's health research, mental healthcare, dementia research, research inclusion, as well as experience working with

communities that are historically underserved by clinical research. Differences in career stage, professional roles, and social positioning informed how the mapping across trial lifecycle stages was interpreted, particularly in identifying where inclusion was not explicitly addressed in existing guidance. Individual perspectives shaped the analysis and interpretation of gaps in inclusion across clinical trial stages.

* Corresponding author. School of Nursing and Midwifery, Aras Moyola, Upper Newcastle Road, University of Galway, Galway, Ireland H91 TK33.

E-mail address: linda.biesty@universityofgalway.ie (L. Biesty).

across the trial lifecycle. This review provides a foundation for developing practical tools to guide more inclusive trials, with the next phase involving coproduction of guiding principles with interest holders. © 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Equity; Diversity; Inclusion; Trial methodology; Trial guidelines; Focused mapping review; Qualitative

1. Introduction

In recent years there has been a notable emphasis on the concepts of equity, diversity and inclusion (EDI) in the context of research practice and design [1]. Omitting research participants from diverse backgrounds makes it challenging to assume the relevance and generalizability of findings to everyday contexts [2]. While important for many research methodologies, this is especially significant to trials aiming to ensure the broader applicability of their results [3].

In response, a growing number of funders, regulatory bodies, and networks, including the National Institutes of Health, Canadian Institute of Health Research, Food and Drug Administration, and Trial Forge, have developed guidance to support more inclusive trial practices [1–4]. However, it remains unclear how well these recommendations align with real-world trial operations, whether they provide adequate guidance across the entire trial lifecycle, or to what extent they are being incorporated in practice.

At the same time, qualitative research, which focuses on understanding people's experiences, the context in which experiences occur, and participant perspectives, has increasingly been recognized as a powerful tool for supporting inclusive research design and evaluation [5,6]. Qualitative research can offer insight into why certain groups remain under-represented, how inclusion strategies and trial recruitment are experienced in practice, and what can be done to adapt trial processes to meet diverse needs. Despite this potential, there is little structured guidance on how qualitative methods can be systematically applied to support EDI in clinical trials [7,8].

The QuAlitative reSearch Supporting IncluSive Trials (Q-ASSIST study) was developed to address this gap by exploring how qualitative methods can support inclusivity across all stages of a clinical trial. This paper presents Phases 1 and 2 of this study, which together contribute guiding principles for inclusive research and offer practical insights for integrating qualitative evidence into clinical trials design methodology. A protocol for this review was registered on the Open Science Framework in July 2024 and is publicly available [9].

2. Aim and objectives

2.1. Aim

To map guidelines and recommendations related to inclusive trials methodology and to identify areas across

these resources where qualitative research could strengthen inclusivity across the trial lifecycle.

2.2. Objectives

- To identify existing guidelines for inclusive trials and trials methodology.
- To identify specific areas within the guidelines where qualitative research can be optimally applied to enhance inclusivity in trials.

3. Methodology

3.1. Phase 1: Focused mapping review and synthesis

3.1.1. Review design: focused mapping review and synthesis

This review was guided by the methodological insights in relation to FMRS offered by Bradbury-Jones et al, following their three-stage approach of (1) focus, (2) mapping, and (3) synthesis [10,11].

An FMRS does not aim to provide an exhaustive systematic review of the literature but instead provides a specific and targeted focus of key features within a research space [12]. This aligns with the purpose of our review and will help generate findings to contribute to the next phases of the Q-ASSIST study.

3.1.2. Guideline identification and selection process: focus

We conceptualized guidelines as nonregulatory recommendations and best practice guidance (eg, frameworks, policy documents) issued by research and academic bodies for the design and conduct of clinical trials. Given the focus of this review and the fact that many guidelines may not be published in peer-reviewed journals, we contained our search to networks/organizations' publication of guidelines supporting inclusive trials methodology that are publicly available to the research community. Specifically, we focused on organizational recommendations rather than published academic papers.

Drawing from the topic expertise of our team (our team includes those with expertise in trial design and feasibility, qualitative research in trials and developing frameworks to support inclusive trials methodology), we searched the publicly available website repositories of national and international institutes and public bodies with established roles in

What is new?

Key findings

- A focused mapping review of 15 publicly available guidelines identified 183 recommendations related to inclusive clinical trial methodology.
- From these, we developed a 12-stage clinical trial framework (QuAlitative reSearch Supporting IncluSive trial stages model) to structure and compare inclusion guidance across the trial lifecycle.
- Mapping against O’Cathain et al’s framework of qualitative research in trials showed that stages such as post-trial access and follow-up, oversight and monitoring, and training and capacity building are under-addressed in both guideline recommendations and prior qualitative research applications.

What this adds to what was known?

- This is the first study to map inclusion-focused clinical trial guidelines using a Standard Protocol Items: Recommendations for Interventional Trials 2025- and Consolidated Standards of Reporting Trials 2025-aligned trial lifecycle framework, and to identify where qualitative research can support inclusion by juxtaposing the findings with the O’Cathain framework of qualitative methods in trials.
- We identified that later trial stages, including implementation, monitoring, and follow-up, lack sufficient guidance on inclusive trials methodology.

What is the implication and what should change now?

- Inclusion efforts should extend beyond recruitment and planning to later stages of the trial lifecycle, including implementation, oversight, and follow-up.
- Trialists and guideline developers should consider targeted use of qualitative methods in under-addressed stages to support equity and strengthen trial delivery.
- These findings inform the next phase of the QuAlitative reSearch Supporting IncluSive Trials study, which will involve coproduction with interest holders, to prioritize gaps and develop guiding principles for inclusive trial design and conduct.

supporting clinical trial design and the conduct of inclusive trials.

Searches were conducted between May 2024 and August 2024 using each organization’s internal website search functions and by manual navigation of guideline, policy, and publication sections. Key terms were applied iteratively to accommodate variation in the websites search functions and indexing and included combinations of terms related to trial design and trials methodology (eg, clinical trial, trial design, trial conduct, trial reporting, recruitment, enrollment); diversity and inclusion (eg, diversity, equity, inclusion, representation, under-represented populations, health equity, health disparities); and trial guidance (eg, guidelines, guidance, recommendations, best practice, frameworks, standards, toolkits) (See [Table A.1](#)).

Identified documents were collated in an Excel document, two reviewers (NO’C, LB) independently applied the following criteria to the identified guidelines to determine those for inclusion in our review. Other members of the review team (LJ, SD) were available to mediate any disagreements and offer insights and opinions.

We included guidelines that fulfill the following criteria:

- Related to the domain of Phase 3 and later phase clinical trials with human participants
- Were publicly available to the research community
- Offered recommendations to researchers in relation to enhancing the design and conduct of inclusive trials.

3.1.3. Mapping process

We mapped each included guideline to a priori data extraction framework. This step also included charting guideline information relating to:

- Country of origin
- Publication date
- Guideline body/organization
- Backgrounds of authors/contributors to the guideline
- Summary of the recommendations
- The underlying evidence to support the recommendations offered in the guideline
- The phases of the trial and trial methods that align with the recommendations of the guideline

We used the PRO-EDI framework (a framework for extracting information about participant characteristics from trials included in a systematic review [13]) to identify the EDI focus of the guideline and if it considers a specific underserved group or if it is broader in its reach. For each recommendation, we recorded which PRO-EDI characteristics were explicitly addressed to understand the frequency and distribution of inclusion focus across guidelines.

This mapping was conducted by one reviewer (NO'C), updated by another reviewer (TT), verified by another member of the team (LB), and reviewed by the team.

3.1.4. Synthesis

The team reviewed each recommendation extracted from the 15 selected guidelines in detail, grouping them by their functional purpose and aligning them with stages of the trial lifecycle. We then conceptually mapped these grouped recommendations to Standard Protocol Items: Recommendations for Interventional Trials 2025 [14] and Consolidated Standards of Reporting Trials 2025 [15] checklist headings to structure the synthesis. The outcome of this process is the Q-ASSIST trial stages model. We then conducted reflexive thematic analysis of all 183 recommendations following Braun and Clarke's six-step approach [16]. TT first familiarized herself with the dataset and generated inductive codes in Excel, capturing explicit and implicit inclusion-related concepts. These codes were then iteratively reviewed, compared, and clustered into preliminary code groups, which were refined through discussion with LB to develop nine higher-order themes. In keeping with reflexive thematic analysis, theme development was an interpretive rather than aggregative process, and final theme labels were shaped to align conceptually with the language of the Q-ASSIST trial stages model. After theme generation, themes were mapped onto the 12 trial stages to examine the distribution of inclusion-focused concepts across the trial lifecycle.

3.2. Phase 2: Revisiting the O'Cathain et al's (2013) framework

3.2.1. Juxtaposing with O'Cathain et al's (2013) framework

To identify which trial stages within the guidelines could be optimally supported by qualitative research, we juxtaposed our Q-ASSIST trial stages model with O'Cathain et al's (2013) framework. O'Cathain's framework provides the most comprehensive, empirically derived synthesis of how qualitative research has been used within randomized controlled trials, drawing on nearly 300 published studies and organized into 22 subcategories ranging from intervention development to recruitment and outcome measures [17,18]. Juxtaposing our mapped stages with this framework enabled us to identify areas where qualitative methods are already used and where qualitative research could strengthen inclusivity but has not yet been routinely applied. This comparison was undertaken as a preparatory step to inform the next phase of the Q-ASSIST study, which will engage trialists, Public and Patient Involvement (PPI) partners, and other stakeholders to explore where qualitative methods could further support EDI-focused trial design and conduct. This phase of synthesis were conducted by TT, verified by LB, and agreed by the wider team.

An overview of the methodological steps undertaken in Phase 1 and 2 of the Q-ASSIST study is presented in Figure 1.

4. Results

4.1. Identification and overview of included trial guidelines

Fifteen guidelines met the inclusion criteria. These guidelines were published between 2001 and 2025 and were developed by national and international research bodies, funding agencies, and collaborative networks. They originated from a range of countries, including the United States (3), United Kingdom (7), Canada (1), Australia (1), Switzerland (1), the Netherlands (1), and international consortia (1), reflecting a diversity of regulatory and research contexts. However, this review did not aim to generate an exhaustive catalog of all existing guidance; guidelines were purposively selected to align with the defined focus of this review and to illuminate how inclusion is addressed across trial methodology.

Across these guidelines, we identified 183 individual recommendations related to inclusive trial design, conduct, reporting, and participant engagement. Details of each guideline, including its name, country of origin, publication date, guiding body/organization are provided in Table A.2.

4.2. Mapping: development of the QuAlitative reSearch Supporting IncluSive trial stages model

The 183 recommendations from the 15 guidelines were reviewed and mapped to a stage of the trial lifecycle based on the functional role it described, regardless of whether the language matched standard trial terminology. Some recommendations explicitly named familiar stages such as "recruitment," "data collection," or "reporting," making the alignment clear. However, others referred more indirectly to activities like building trust, ensuring participant support, or engaging communities—concepts that required interpretive judgment to determine their alignment with trial processes.

As more recommendations were mapped, it became evident that while the concepts were consistent, the language varied significantly across guidelines. To address this, we consolidated overlapping concepts and aligned them with Standard Protocol Items: Recommendations for Interventional Trials 2025 and Consolidated Standards of Reporting Trials 2025 checklists (See Table A.3). We reviewed both checklists side by side to identify overlapping domains and organized similar items into broader categories that matched the real-life flow of a trial, from planning to reporting. This process resulted in a 12-stage methodology framework—the Q-ASSIST trial stages

model (Table 1)—that reflects the full trial lifecycle and the areas where inclusion is addressed.

As part of the mapping process, each trial stage was reviewed across each individual guideline to determine whether it was explicitly addressed, not applicable, or represented by a gap. If a guideline scope clearly did not intend to cover a particular methodological stage, such as post-trial care in a design-only guideline, it was marked as not applicable. Conversely, if a relevant stage fell within the intended scope of the guideline but lacked corresponding recommendations, it was marked as a gap. This distinction allowed us to avoid overinterpreting absence as omission and enabled a more accurate reflection of both coverage and limitations across diverse types of guidelines.

4.3. Mapping to clinical trial methodology stages

The review of 15 guidelines revealed varied coverage across the Q-ASSIST trial stages model (Fig 2). Early stages such as population definition and sampling strategy;

trial preparation and planning; and ethical review, consent and governance were addressed in most guidelines. Later stages like trial implementation and fidelity; data analysis and interpretation; and post-trial access and participant follow-up were less consistently covered.

4.4. Coverage of PRO equity, diversity and inclusion characteristics

Guideline recommendations showed varied emphasis across PRO-EDI characteristics (Figs 3 and 4). Guidelines frequently addressed socioeconomic status, race/ethnicity, gender, and disability. Location, sex, and age were moderately represented. Sexual identity and education were least covered. Only 4 of 15 guidelines demonstrated any implicit attention to intersectionality, and none offered explicit or operational intersectional guidance. Details of the intersectionality coverage across the 15 guidelines are provided in Table A.4.

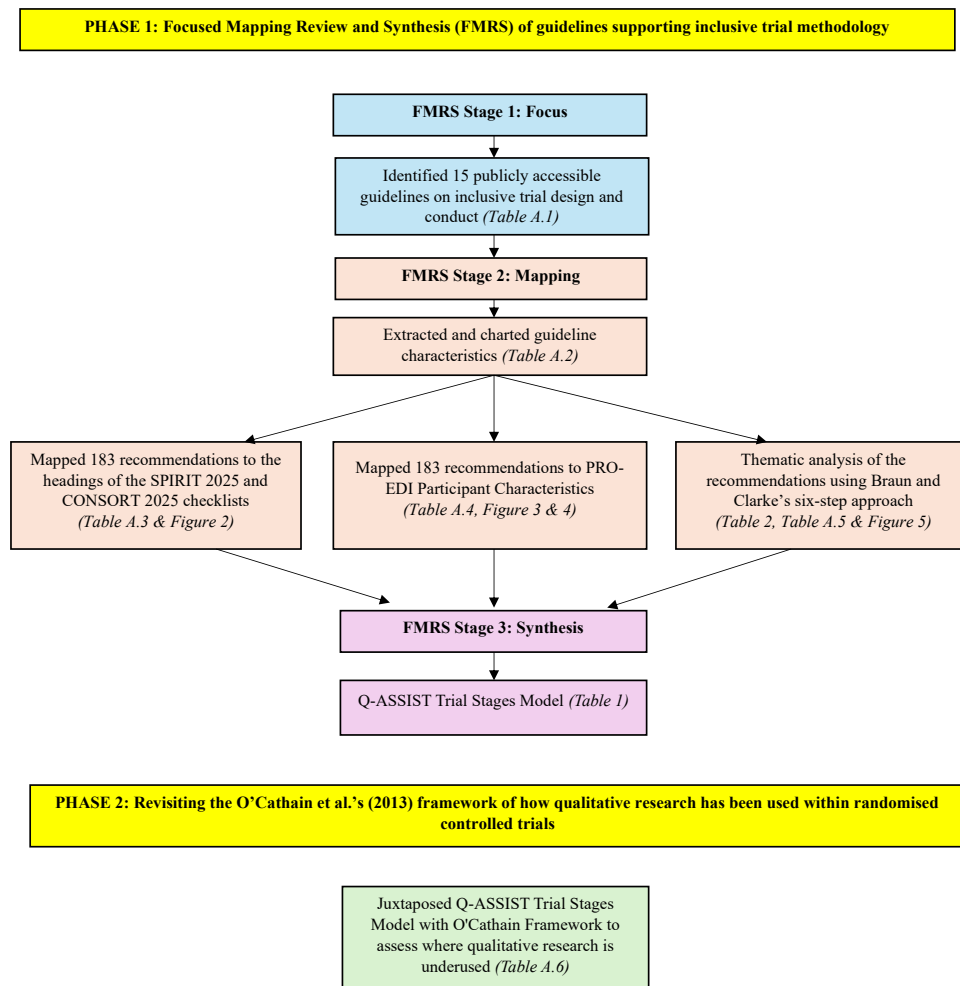


Figure 1. Schematic overview of Phase 1 and Phase 2 methodological steps in the Q-ASSIST study. FMRS, focused mapping review and synthesis; Q-ASSIST, QuAlitative reSearch Supporting IncluSive Trial; EDI, equity, diversity, and inclusion; SPIRIT, Standard Protocol Items: Recommendations for Interventional Trials; CONSORT, Consolidated Standards of Reporting Trials. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.).

4.5. Thematic coverage across guidelines

Nine cross-cutting inclusion-focused concepts (Table 2) were interpreted through thematic analysis and mapped to the Q-ASSIST trial stages model (Fig 5). The results show that most guidelines focus heavily on the early stages of the trial process. In contrast, several later stages in the trial lifecycle were supported by fewer themes. For example, post-trial access and participant follow-up; training and capacity building; and trial oversight and monitoring received limited attention. Notably, trial implementation and fidelity and data analysis and interpretation were not directly addressed by any of the identified themes. A full table of the themes developed through our thematic analysis, along with representative codes and refined theme labels that reflects its focus and terminology alignment with Q-ASSIST trial stages model, is provided in Table A.5.

4.6. O’Cathain framework mapping

The juxtaposition of our Q-ASSIST trial stages model with O’Cathain et al’s (2013) framework highlighted that qualitative research is most often considered relevant in early to midtrial stages. However, several later trial stages that we identified as important for inclusive trials methodology through our mapping and synthesis—specifically post-trial access and participant follow-up, trial oversight and monitoring, and training and capacity building—are in addition to the stages noted in O’Cathain’s framework (See Table A.6).

5. Discussion

This FMRS mapped existing guidelines for inclusive clinical trials to a 12-stage methodology framework—the Q-ASSIST trial stages model—to identify the potential contribution qualitative research can make in enhancing EDI across the trial lifecycle. The mapping process showed that most guidelines focus on inclusion-related considerations during early trial stages such as planning, population definition, and ethical review but offer less guidance on later stages such as trial implementation, data analysis, oversight, and post-trial care. This shows a gap and points to the need for clearer guidance that supports inclusion at every stage of the process.

5.1. Mapping of guideline recommendations to clinical trial methodology stages and thematic groupings

Across guidelines, inclusion considerations were heavily concentrated in early trial stages. These findings are consistent with broader literature indicating that most inclusion efforts focus on getting participants into trials, with far less attention to supporting inclusion during implementation, monitoring, analysis, or follow-up [3,8,19]. By organizing recommendations across 12 clearly defined trial stages,

our FMRS extends this literature by showing precisely where these gaps occur, rather than only describing them generically.

Our thematic analysis echoed these findings. Without stronger and more explicit guidance on inclusion in the later stages of trial, there is a risk that inclusive intentions established during planning may not be maintained throughout the trial lifecycle. For example, a trial might recruit a diverse sample but fail to analyze data in ways that reflect subgroup differences or return results in formats that are accessible to all participants. Our findings reinforce the need for future guidelines to provide more practical, stage-specific strategies to sustain inclusive practice throughout the entire trial lifecycle.

5.2. Mapping of guideline recommendations to PRO equity, diversity and inclusion characteristics

Mapping to PRO-EDI characteristics showed that many guidelines used broad terms such as “diverse” or “underserved,” limiting clarity on which groups were intended. These omissions are consistent with broader concerns in EDI scholarship, which note that single-axis focus can hide the lived realities of people who sit at the intersection of several forms of disadvantage [20] and can create blind spots in implementation. Without explicit attention to characteristics such as sexual identity, education, and language or cultural identity, trial processes risk reproducing inequities even when demographic diversity is achieved at recruitment. Our findings highlight that intersectional and context specific understandings of inclusion are rarely embedded in operational guidance, which may limit the ability of trialists to meaningfully support equity across the lifecycle of a trial.

5.3. Mapping of O’Cathain’s framework

Juxtaposing our findings to O’Cathain et al’s framework showed that qualitative methods are primarily used in early to midtrial stages, with limited integration in oversight, training, or post-trial follow-up. This mirrors the gaps identified in our guideline mapping and suggests a missed opportunity: qualitative research could support inclusive implementation, responsive monitoring, and participant-centered post-trial processes, yet is rarely deployed in these areas. Our mapping therefore helps identify where methodological guidance could evolve to embed qualitative methods more comprehensively.

5.4. Implications

This FMRS highlights patterns in how inclusion is addressed across guidelines and where conceptual gaps persist. Juxtaposing our findings with O’Cathain et al’s framework helps clarify which parts of the trial lifecycle already align with established uses of qualitative research, and which areas receive little attention. This comparison

Table 1. The Q-ASSIST trial stages model: summary of the 12 trial stages with definitions and stage specific EDI considerations generated from the focused mapping review of 15 guidelines

Number	Stage name	Description	Equity, diversity and inclusion considerations
1	Population definition and sampling strategy	How trial populations are defined, and sampling strategies are identified	EDI considerations include avoiding unjustified exclusions, enabling representation of underserved groups, and planning sampling strategies (eg, oversampling, stratification) that support equity and subgroup analyses
2	Trial preparation and planning	Activities involved in designing the trial and setting up the infrastructure needed to run it	EDI considerations include early planning for inclusive design, involving diverse expertise, and anticipating structural barriers that may influence participation
3	Ethical review, consent and governance	Processes for gaining ethical approval, consent procedures, and regulatory compliance	EDI considerations include culturally responsive consent processes, adaptations for varying literacy or capacity levels, and governance mechanisms that respect community data rights and autonomy
4	Community engagement	Building trust and collaboration with communities involved in or affected by the trial	EDI considerations include early and sustained engagement, codevelopment of procedures with communities, and ensuring cultural appropriateness in communication and decision-making
5	Recruitment, support and retention	Efforts to recruit and retain diverse participants and support their trial experience	EDI considerations include culturally sensitive recruitment strategies, reducing participation burdens (eg, transport, childcare), and ensuring supportive, respectful interactions throughout participation
6	Data collection and quality management	How data are collected, managed, and the steps taken to assure quality and consistency of the processes	EDI considerations include ensuring that questionnaires, scales, and data collection tools are valid across different ages, cultures, literacy levels, and languages; capturing data that supports subgroup analysis so that differences between demographic groups are not overlooked; and applying culturally safe data management practices that respect privacy, Indigenous data sovereignty, and community governance
7	Trial implementation and fidelity	Ensuring interventions are delivered as intended and trial processes are followed	EDI considerations include monitoring equitable delivery of interventions, identifying contextual barriers, and ensuring that implementation does not disadvantage particular groups
8	Trial oversight and monitoring	Monitoring of trial progress, safety, and fidelity to protocols	EDI considerations include ensuring diverse representation in oversight bodies (eg, Data and Safety Monitoring Boards) and monitoring for differential harms or burdens across participant subgroups
9	Training and capacity building	Training and ongoing support for trial staff and others involved in delivering or supporting the trial (e., local site teams, PPI contributors, community organizations).	EDI considerations include cultural competence training, strengthening capacity of under-resourced teams, and enabling meaningful involvement of community representatives
10	Data analysis and interpretation	How data are analyzed and interpreted (including transparent subgroup reporting when appropriate)	EDI considerations include transparent subgroup analysis, and interpreting findings within relevant sociocultural contexts
11	Post-trial access and participant follow-up	Ongoing care, communication of results, access to interventions, and follow-up to understand participants' experiences after the trial ends	EDI considerations include equitable access to post-trial interventions, culturally appropriate dissemination of results, and recognition of different participant needs after trial completion
12	Reporting and knowledge translation	Dissemination of results and translation of findings into practice and policy	EDI considerations include accessible dissemination materials (eg, plain language summaries), returning results to communities, and transparently reporting inclusion practices and outcomes

EDI, equity, diversity, and inclusion; PPI, Public and Patient Involvement; Q-ASSIST, QuAlitative reSearch Supporting IncluSive Trial.

makes it easier to see where inclusion is currently supported within trials methodology and where gaps remain.

For guideline developers and funders, the findings provide a template for clarifying the scope, intentionality, and specificity of future inclusion recommendations. For

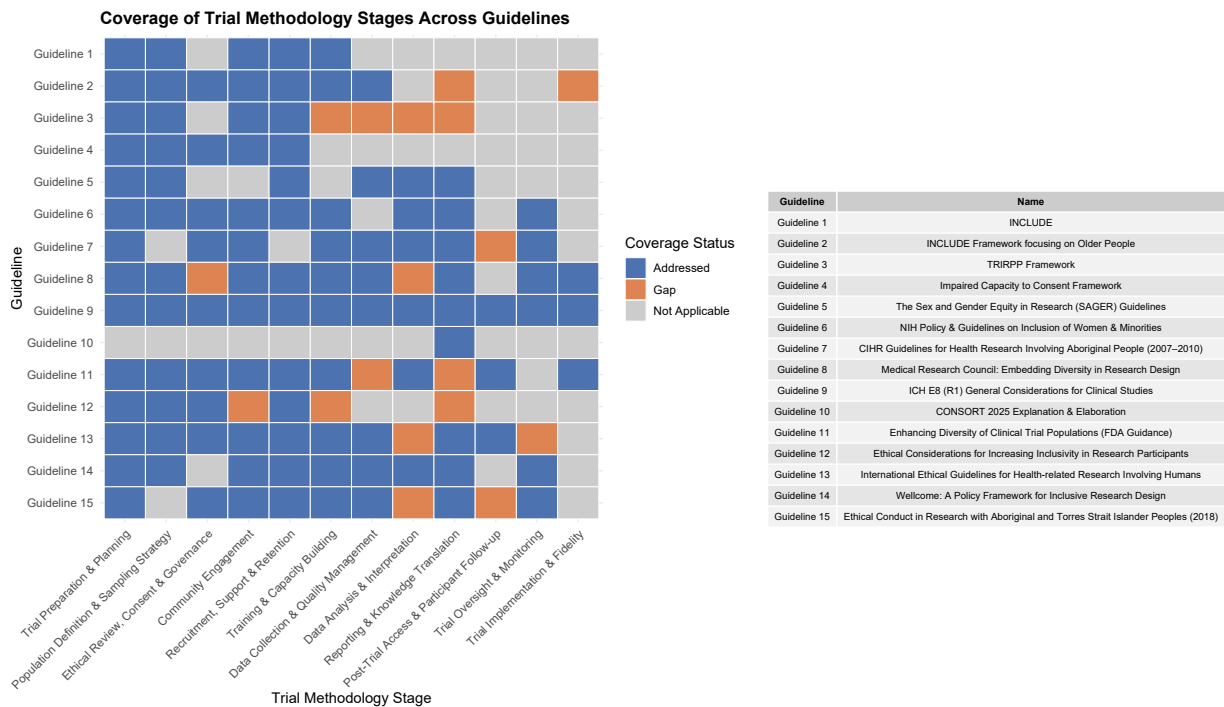


Figure 2. Mapping clinical trial guidelines to the Q-ASSIST trial stages model. Heatmap illustrating the extent to which each of the Q-ASSIST trial stages model are addressed, represented a gap, or was not applicable to that guideline. Q-ASSIST, QuAlitative reSearch Supporting IncluSive Trial; CONSORT, Consolidated Standards of Reporting Trials' NIH, National Institutes of Health; CIHR, Canadian Institute of Health Research; FDA, Food and Drug Administration; INCLUDE, Innovations in Clinical Trial Design and Delivery for the Under-served; TRIRPP, the trauma and resilience informed research principles and practices framework. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.).

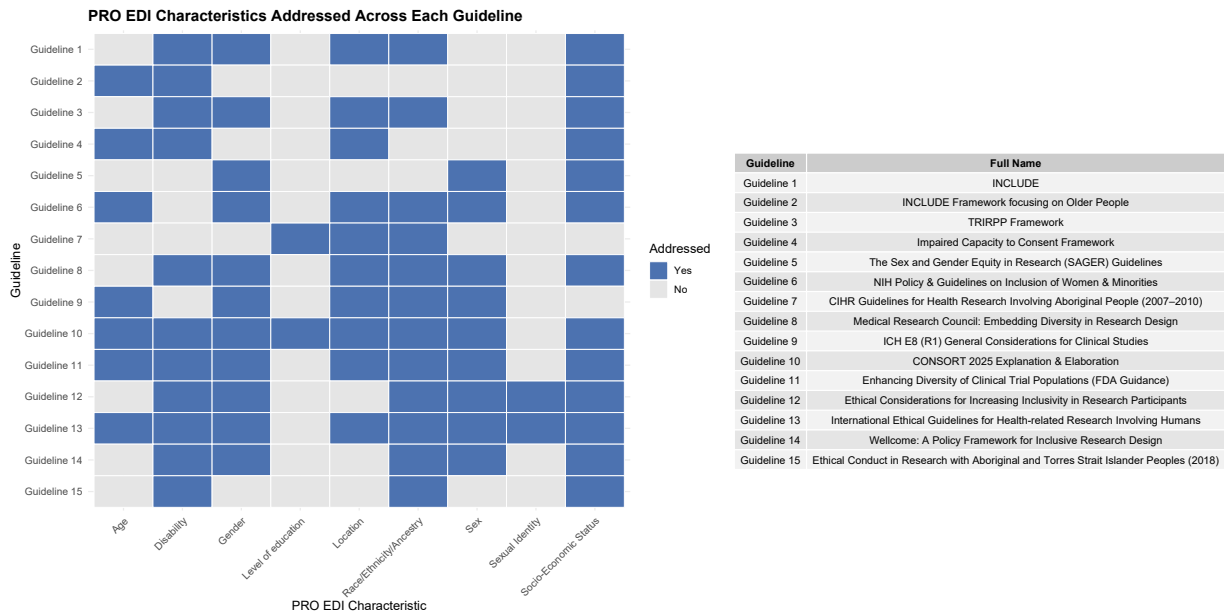


Figure 3. PRO-EDI characteristics addressed across 15 guidelines. Presence or absence of each PRO-EDI characteristic in guideline recommendations. EDI, equity, diversity, and inclusion; NIH, National Institutes of Health; CIHR, Canadian Institute of Health Research; FDA, Food and Drug Administration; INCLUDE, Innovations in Clinical Trial Design and Delivery for the Under-served; TRIRPP, the trauma and resilience informed research principles and practices framework. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.).

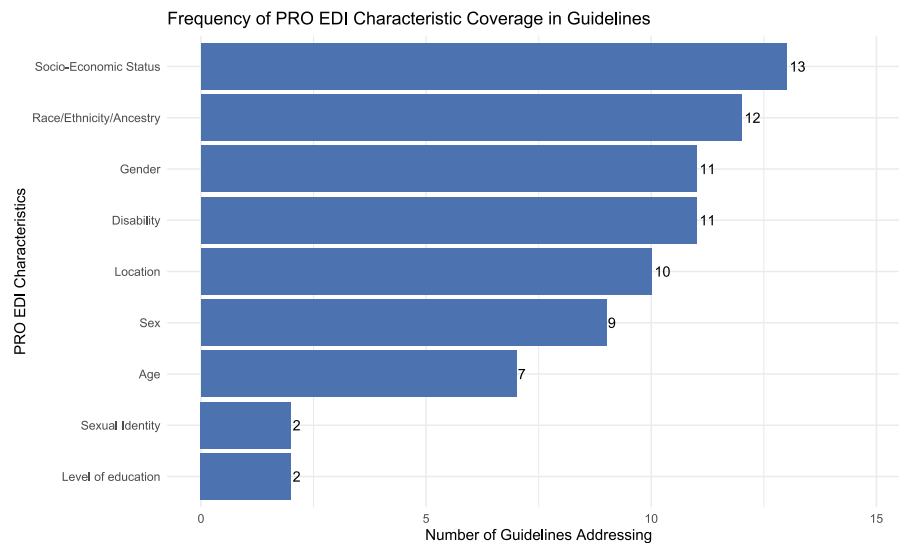


Figure 4. Frequency of PRO-EDI characteristic coverage in guidelines. Number of guidelines addressing each PRO-EDI characteristic, illustrating overall coverage patterns across the dataset. EDI, equity, diversity, and inclusion. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.).

trialists, it emphasizes the value of adopting lifecycle-wide planning tools that integrate both EDI and qualitative methods into routine trial processes. These implications inform Phase 3 of the Q-ASSIST study, where coproduction activities will translate these insights into practical guidance and priority areas for action.

6. Limitations of this study

This study has limitations that should be acknowledged. We focused exclusively on publicly available, English-

language guidelines, which may have excluded relevant documents developed in other languages or regional contexts. Drawing on the topic expertise of our team, the review focused on guidance issued by a defined group of organizations (as noted in 3.2), all of which are based in high-income countries. As a result, relevant English-language guidance developed outside this organizational scope (eg, by other international bodies or regional initiatives) may not have been captured.

While we aimed to capture the original intent of each guideline as accurately as possible, interpretive judgment was required when mapping vague or variably phrased

Table 2. Summary of the nine themes generated through reflexive thematic analysis, with brief descriptions of each theme

Number of theme	Theme	Brief description (what the theme represents)
1	Inclusive trial design and sampling strategy	Focuses on designing trials that ensure representative participation through inclusive sampling strategies, justified eligibility decisions, and attention to structural barriers from the outset.
2	Equitable eligibility and recruitment strategy	Addresses fair recruitment practices, avoidance of unjustified exclusions, equitable burden-benefit distribution, reimbursement, and inclusion of groups often excluded from research.
3	Community engagement and governance	Emphasizes respectful, continuous engagement with communities, including indigenous or local governance structures, shared decision-making, and community control over elements of trial conduct.
4	Culturally responsive consent and ethics	Captures the need for culturally appropriate consent processes, continuous consent, adaptations for literacy and cultural norms, and ethical review processes that uphold participant autonomy.
5	Cultural competency in community engagement	Highlights the importance of cultural capability, training, and respect for Indigenous knowledge systems to support inclusive, context-sensitive engagement.
6	Ethical data use and community ownership	Reflects community rights to data ownership, governance of biological samples, sovereignty over knowledge, and agreements ensuring ethical, community-led data use.
7	Transparent reporting and trial accountability	Focuses on transparency in ethics, training, reporting results back to communities, and clear accountability mechanisms throughout the trial lifecycle.
8	Capacity building for inclusive research	Addresses training, education, and strengthening local and community-based research and ethics capacity to support ongoing inclusive research practices.
9	Risk mitigation and participant safeguarding	Covers minimizing risk, assessing differential impacts on groups, protecting participants in outbreak/disaster settings, and ensuring ongoing monitoring and responsive mitigation.

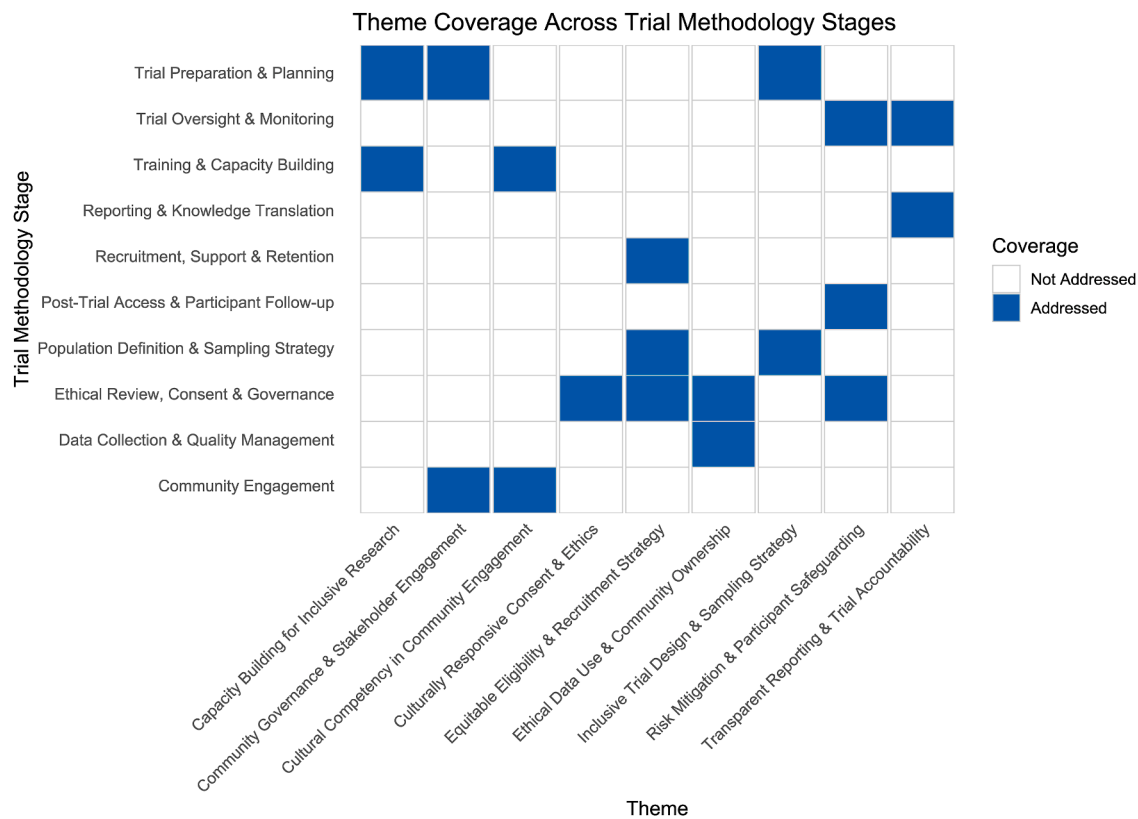


Figure 5. Theme coverage across trial methodology stages. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.).

recommendations to specific trial stages or PRO-EDI characteristics. Methodologically, we selected a review typology that offered specific, targeted focus resulting in a narrow scope of sources; the included guidelines do not constitute an exhaustive list. While this approach aligns with our current research aims, we acknowledge that future research in this area may benefit from a systematic scoping review.

7. Conclusion

This FMRS provides a comprehensive view of how inclusion-focused clinical trial guidelines currently address different stages of the trial lifecycle. Our findings demonstrate that most guidance is concentrated in the early stages of trial design and planning, with limited attention to later stages such as implementation, monitoring, and post-trial engagement. This highlights a critical need for more comprehensive guidance to ensure inclusive practice is maintained throughout the stages of a trial. In exploring how qualitative research could help address these gaps, we found that while qualitative methods are commonly applied in early trial phases, their use in oversight, training, and post-trial care remains limited indicating an opportunity for greater integration in under-addressed stages.

Although PPI was not included in this initial phase, the broader Q-ASSIST study will have substantial PPI

involvement. All interest holders, including PPI contributors, will be engaged in coproduction activities to use to the findings of this review to coproduce practical guidance for how qualitative research can support inclusive practice at each trial stage.

CRediT authorship contribution statement

Teena Thomas: Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Data curation. **Laura Jones:** Writing – review & editing, Validation, Methodology, Conceptualization. **Shoba Dawson:** Writing – review & editing, Validation, Methodology, Conceptualization. **Catherine Houghton:** Writing – review & editing, Funding acquisition, Conceptualization. **Andrew Hunter:** Writing – review & editing, Funding acquisition, Conceptualization. **Niamh O’Connell:** Writing – review & editing, Formal analysis. **Linda Biesty:** Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

TT received financial support from Health Research Board- Trials Methodology Research Network. SD is a

member of the Representation in Global Clinical Trials Advisory Board for Boehringer and Ingelheim.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112202>.

Data availability

Data will be made available on request.

References

- [1] Kelsey MD, Patrick-Lake B, Abdulai R, Broedl UC, Brown A, Cohn E, et al. Inclusion and diversity in clinical trials: actionable steps to drive lasting change. *Contemp Clin Trials* 2022;116: 106740. <https://doi.org/10.1016/j.cct.2022.106740>.
- [2] Treweek S, Banister K, Bower P, Cotton S, Devane D, Gardner HR, et al. Developing the INCLUDE Ethnicity Framework: a tool to help trialists design trials that better reflect the communities they serve. *Trials* 2021;22(1):337. <https://doi.org/10.1186/s13063-021-05276-8>.
- [3] Bodicoat DH, Routen AC, Willis A, Ekezie W, Gillies C, Lawson C, et al. Promoting inclusion in clinical trials: a rapid review of the literature and recommendations for action. *Trials* 2021;22(1):880. <https://doi.org/10.1186/s13063-021-05849-7>.
- [4] Dawson S, Banister K, Biggs K, Cotton S, Devane D, Gardner H, et al. Trial Forge Guidance 3: randomised trials and how to recruit and retain individuals from ethnic minority groups—practical guidance to support better practice. *Trials* 2022;23(1):672. <https://doi.org/10.1186/s13063-022-06553-w>.
- [5] Stockless A, Brière S. How to encourage inclusion in a qualitative research project using a design-based research methodology. *Int J Qual Methods* 2024;23:1–9. <https://doi.org/10.1177/16094069241227852>.
- [6] Renjith V, Yesodharan R, Noronha JA, Ladd E, George A. Qualitative methods in health care research. *Int J Prev Med* 2021;12(20):1–8. https://doi.org/10.4103/ijpvm.IJPVM_321_19.
- [7] Evans C, Hassanein ZM, Bains M, Bennett C, Bjerrum M, Edgley A, et al. Addressing equity, diversity, and inclusion in JBI qualitative systematic reviews: a methodological scoping review. *JBI Evid Synth* 2025;23(3):454–79. <https://doi.org/10.11124/jbies-24-00025>.
- [8] Biggs K, Hullock K, Dix C, Lane JA, Green H, Treweek S, et al. Time to STEP UP: methods and findings from the development of guidance to help researchers design inclusive clinical trials. *BMC Med Res Methodol* 2024;24:227. <https://doi.org/10.1186/s12874-024-02342-y>.
- [9] O'Connell N, Jones L, Dawson S, Houghton C, Hunter A, Biesty L. Existing research guidelines for inclusive trial methodology: working toward the intersection of qualitative research, EDI and trial methodology: a focused mapping review protocol. *OSF Preprints* 2024. <https://doi.org/10.17605/OSF.IO/MA4NF>.
- [10] Bradbury-Jones C, Breckenridge J, Clark MT, Herber OR, Wagstaff C, Taylor J. The state of qualitative research in health and social science literature: a focused mapping review and synthesis. *Int J Soc Res Methodol* 2017;20:627–45. <https://doi.org/10.1080/13645579.2016.1270583>.
- [11] Bradbury-Jones C, Breckenridge J, Clark MT, Herber OR, Jones C, Taylor J. Advancing the science of literature reviewing in social research: the focused mapping review and synthesis. *Int J Soc Res Methodol* 2019; 22:451–62. <https://doi.org/10.1080/13645579.2019.1576328>.
- [12] Croft E, Clark MT, Efstathiou N, Bradbury-Jones C. A focused mapping review and synthesis of a priori risk factors associated with medical misconduct. *BMJ Open Qual* 2019;8:e000538. <https://doi.org/10.1136/bmjopen-2018-000538>.
- [13] Trial Forge. PRO-EDI participant characteristics table. Available at: <https://www.trialforge.org/trial-diversity/pro-edi-improving-how-equity-diversity-and-inclusion-is-handled-in-evidence-synthesis/>. Accessed March 18, 2025.
- [14] Chan AW, Boutron I, Hopewell S, Moher D, Schulz KF, Collins GS, et al. SPIRIT 2025 statement: updated guideline for protocols of randomised trials. *BMJ* 2025;389:e081477. <https://doi.org/10.1136/bmj-2024-081477>.
- [15] Hopewell S, Chan AW, Collins GS, Hróbjartsson A, Moher D, Schulz KF, et al. CONSORT 2025 explanation and elaboration: updated guideline for reporting randomised trials. *BMJ* 2025;389: e081124. <https://doi.org/10.1136/bmj-2024-081124>.
- [16] Braun V, Clarke V. Using thematic analysis in psychology. *Qual Res Psychol* 2006;3(2):77–101.
- [17] O'Cathain A, Thomas KJ, Drabble SJ, Rudolph A, Hewison J. What can qualitative research do for randomised controlled trials? A systematic mapping review. *BMJ Open* 2013;3(6):e002889. <https://doi.org/10.1136/bmjopen-2013-002889>.
- [18] O'Cathain A, Thomas KJ, Drabble SJ, Rudolph A, Goode J, Hewison J. Maximising the value of combining qualitative research and randomised controlled trials in health research: the QUALitative Research in Trials (QUART) study—a mixed methods study. *Health Technol Assess* 2014;18:1–197.
- [19] Garavito GA, Gilchrist K, Ciurtin C, Khanna S, Chambers P, McNally N, et al. Enablers and barriers of clinical trial participation in adult patients from minority ethnic groups: a systematic review. *Trials* 2025;26:65. <https://doi.org/10.1186/s13063-025-0065>.
- [20] Bamba C. Placing intersectional inequalities in health. *Health Place* 2022;78:102761. <https://doi.org/10.1016/j.healthplace.2022.102761>.